

Stratification of viral shedding patterns in saliva of COVID-19 patients

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This **important** work attempts to understand observed variability in oral shedding of SARS-CoV-2 and suggests that routine clinical factors are not determinative. The evidence supporting the conclusion is **solid** though the limited clinical heterogeneity of the included cohorts, the lack of COVID vaccination, and the absence of comprehensive viral load data for model training, makes the results difficult to generalize to contemporaneous COVID-19 conditions. This study may be of interest to virologists, public health officials and clinicians.

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Abstract

Living with COVID-19 requires continued vigilance against the spread and emergence of variants of concern (VOCs). Rapid and accurate saliva diagnostic testing, alongside basic public health responses, is a viable option contributing to effective transmission control. Nevertheless, our knowledge regarding the dynamics of SARS-CoV-2 infection in saliva is not

as advanced as our understanding of the respiratory tract. Here we analyzed longitudinal viral load data of SARS-CoV-2 in saliva samples from 144 patients with mild COVID-19 (a combination of our collected data and published data). Using a mathematical model, we quantified individual-level viral dynamics and stratified them into three groups using a clustering approach. Notably, the three groups exhibited distinct differences viral RNA detection durations: 11.5 days (95% CI: 10.6 to 12.4), 17.4 days (16.6 to 18.2), and 30.0 days (28.1 to 31.8), respectively. Surprisingly, this stratified grouping remained unexplained despite our analysis of 47 types of clinical data, including basic demographic information, clinical symptoms, results of blood tests, and vital signs. Additionally, we quantified the expression levels of 92 micro-RNAs in a subset of saliva samples, but these also failed to explain the observed stratification, although the mir-1846 level may have been weakly correlated with peak viral load. Our study provides insights into SARS-CoV-2 infection dynamics in saliva, highlighting the challenges in predicting the duration of viral RNA detection without indicators that directly reflect an individual's immune response, such as antibody induction. Given the significant individual heterogeneity in the kinetics of saliva viral shedding, identifying biomarker(s) for viral shedding patterns will be crucial for improving public health interventions in the era of living with COVID-19.

Main Text

Coronavirus disease 2019 (COVID-19) vaccinations, which are effective in preventing infection by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) and severe COVID-19 illness, have enabled the gradual and safe removal of COVID-19 restrictions on everyday life over the past year. However, we still face the emergence of variants of concerns (VOCs), which is a major worry in the era of “living with COVID-19”. As a result, to prevent major outbreaks, basic public health responses such as testing, isolation, and quarantine are demanded and important. In particular, rapid and accurate diagnostic tests are essential for controlling ongoing transmission. Salivary diagnostic testing is a convenient tool for early and efficient diagnosis of COVID-19 because it is easy for health care professionals and patients to administer.^{1,2}

The oral cavity is an important target for SARS-CoV-2³, and viral particles in the lower and upper respiratory tract can reach the oral cavity through liquid droplets.¹ Saliva droplets are thus a potential route of SARS-CoV-2 transmission.^{1,3} Although SARS-CoV-2 infection dynamics within the upper respiratory tract are well characterized thanks to data from oropharyngeal and nasopharyngeal swabs⁴⁻⁹, infection dynamics in saliva are poorly understood. One potential explanation is the practical challenges inherent in using saliva as a specimen type. Handling steps such as tube opening, pipetting, and vortexing can generate infectious aerosols, requiring strict biosafety precautions (e.g., work in a biosafety cabinet). Also, salivary RNases and the abundant oral microbiota may degrade viral RNA or increase background noise, particularly in low viral load samples.

Recent reports^{3,5} have suggested that SARS-CoV-2 infection dynamics differ qualitatively across tissues. Salivary gland tissue (parotid, submandibular, and sublingual) and other tissues are highly compartmentalized, and saliva represents the corresponding biofluid of the oral compartment. However, although individual-level heterogeneity in virus dynamics (especially individual infectiousness) has been evaluated, it remains unknown how individual-level viral shedding patterns in saliva are stratified, and which factor(s) determine the patterns.^{5,10} Since saliva is a route of direct viral transmission, salivary viral load provides particularly valuable information for assessing a patient's immediate infectiousness compared with measurements from other anatomical sites. Accordingly, to shape a country's early response to future VOCs, such as through isolation and screening guidelines based on salivary diagnostic testing, it is critical to address these points with the use of a high quality and quantity of saliva specimens annotated with basic clinical patient data.

So far, our ability to understand and characterize whole SARS-CoV-2 infection dynamics has been hindered by several limitations of the clinical data that made it impossible to capture either the early or the late phase of infection or to annotate individual-level clinical information. To overcome these limitations, here we quantified and stratified longitudinal virus dynamics in saliva samples from 144 mildly symptomatic participants of two different but complementary cohorts^{5,11,12}. We successfully identified three groups with distinct viral shedding patterns. Notably, the groups showed clear differences in the duration of viral RNA detection (the mean durations were 11.5 days, 17.4 days, and 30.0 days, respectively). These findings imply a large inter-participant heterogeneity in virus infection dynamics. However, when we analyzed a total of 47 variables, including basic demographic information, daily clinical symptoms, results of blood tests, and vital signs, none explained the stratified grouping.

We also retrospectively explored whether salivary micro-RNAs were associated with the stratification by using stored residual saliva specimens. Micro-RNAs are non-coding RNAs that regulate numerous cellular processes by modulating protein levels through direct binding to mRNA (coding-RNA), thereby influencing translation efficiency or mRNA abundance. Various microRNAs are recognized for their impact on the viral replication ability and immune response in viruses like EBV, HCV, and HIV etc.¹³. Many studies are underway to investigate micro-RNAs as potential targets for virus diagnosis and treatment. However, the relationship between micro-RNAs and the patterns of virus shedding of the SARS-CoV-2 virus in body remains unknown. We quantified the expression levels of 92 micro-RNAs and found that no micro-RNA significantly explained the stratified groups, although the mir-1846 level may have been weakly correlated with the peak viral load. Our findings provide important insights into the complexities of viral shedding patterns in saliva and suggest that predicting the heterogeneity of viral dynamics using basic clinical and micro-RNA data may be a challenging task. These insights are critical for developing accurate diagnostic tools, effective treatments, and prevention strategies for COVID-19.

Results

Description of cohort and study design

We used longitudinal saliva viral load data obtained from cohorts of the nelfinavir (NFV) clinical trial (JRCT2071200023^{11,12}) and the University of Illinois at Urbana-Champaign⁵. All infections were either mild or asymptomatic. All participants in these cohorts reported through surveys (self-declaration) that they had never been previously infected with SARS-CoV-2, and none were vaccinated against SARS-CoV-2 at the time of enrolment (see Methods for details, including the measurement dates). Of 182 participants from these two studies, 144 symptomatic participants, excluding 22 asymptomatic participants and 16 participants with incomplete observational data, were considered in our analysis (**Fig 1A**). In addition, we annotated the saliva viral load results of 90 participants from the NFV clinical trial with their sex, age, daily symptoms, blood test results, and vital signs. These clinical data were collected longitudinally for each patient (**Fig 1B**). While all patients had clinical measurements at the enrollment time point (a mean of 2.9 days after symptom onset), the frequency of subsequent measurements varied among individuals (**Fig 1B**). Therefore, our analysis focused on the clinical data obtained at enrollment, which were consistently available across all patients. This approach is also appropriate given the study's aim of identifying early predictors of viral dynamics (see later).

Table 1 and **Supplementary Table 1** present the mean values of clinical information and daily symptom data, respectively. The clinical data described in **Table 1** were collected only for the NFV cohort. Therefore, we first used both cohorts to estimate viral dynamics, but in the subsequent analyses examining the relationship between the clinical data and estimated viral

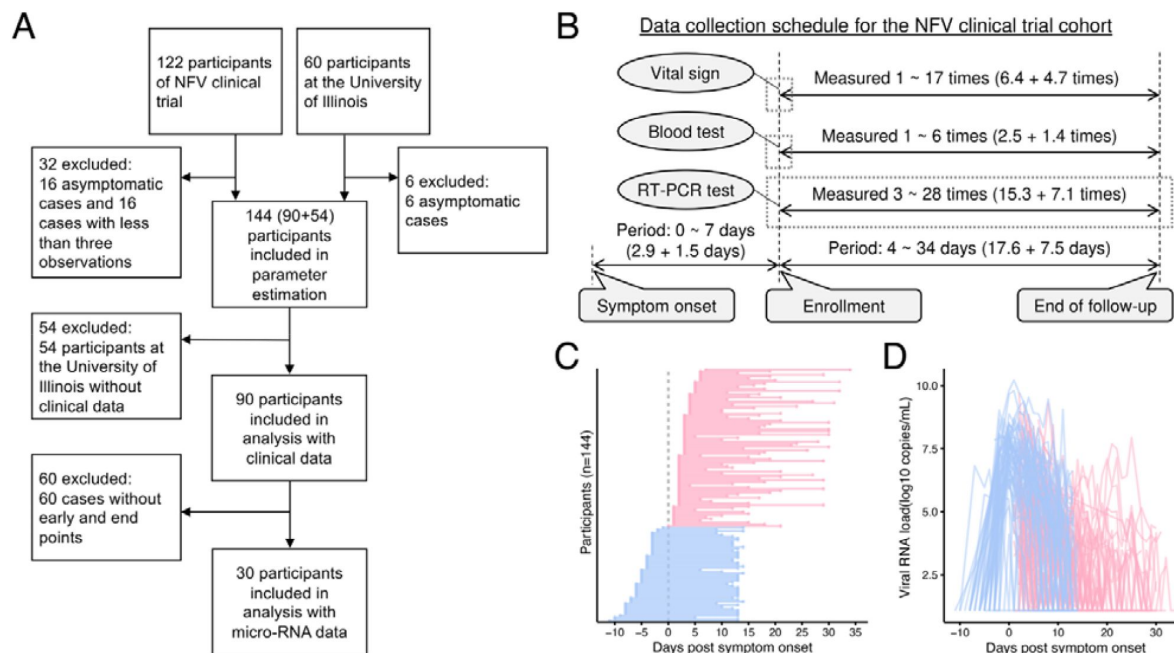


Figure 1.

Characteristics of cohorts from the NFV clinical trial (jRCT2071200023¹¹¹²) and the University of Illinois⁵:

(A) Flowchart of cohorts from the NFV clinical trial and the University of Illinois along with the number of participants and inclusion criteria for our analysis. **(B)** Data collection schedule of viral load, blood test, and vital signs from participants in the NFV clinical trial is described. The grey box highlights the measurement dates for the observations included in the analysis. **(C)** and **(D)** show, for each participant (N=144 participants, 2191 samples), the timeline of sample collection and the captured SARS-CoV-2 viral RNA load for saliva RT-qPCR, respectively. The red and blue colors indicate samples for cohorts from the NFV clinical trial and the University of Illinois, respectively.

dynamics, we relied only on the NFV cohort. Because daily symptom data described in **Supplementary Table 1** were collected for both cohorts, both cohorts were included in the analyses relating daily symptom data to viral dynamics.

The data from the two studies complemented one another: the longitudinal data from the University of Illinois contained data on the early phase of infection (before symptom onset), while the data from the NFV clinical trial contained data on the late phase of viral RNA load (more than 14 days after symptom onset) (**Fig 1C**). Viral load data were collected an average of 15.2 times per participant (SD, 5.7). Pre-symptom onset measurements averaged 0 (SD, 0) in the NFV cohort and 3 (SD, 1) in the Illinois cohort. At 14 or more days after symptom onset, the corresponding averages were 5.5 (SD, 6.1) and 0.1 (SD, 0.3), respectively. The time-series viral load in the saliva samples for all individual used in the analysis are plotted in **Fig 1D**. We also analyzed 60 stored saliva specimens from 30 participants of the NFV clinical trial for micro-RNA analysis.

Quantifying and stratifying SARS-CoV-2 infection dynamics in saliva

We employed a previously developed mathematical model describing SARS-CoV-2 infection dynamics [i.e., Eqs.(1-2)] to evaluate interparticipant heterogeneity (details are provided in Methods), and reconstructed the best-fit virus dynamics in saliva of 144 symptomatic participants (**Supplementary Fig 1** and **Supplementary Table 2**). We also applied a mechanistically more realistic mathematical model Eqs.(3-6) that we had previously used with the Illinois dataset⁵ (**Supplementary Table 3**). Both models showed stable results in the visual predictive checks (VPC) and convergence assessments and adequately described the data (**Supplementary Fig 2** and **Supplementary Fig 3**). The fit with Eqs.(3-6) yielded a lower Akaike Information Criterion (AIC) of 6,720, compared with 6,834 obtained with Eqs.(1-2). Nevertheless, the reconstructed viral dynamics in the NFV clinical trial dataset were highly similar between the two models, whereas somewhat larger differences were observed for the Illinois dataset (**Supplementary Fig 4**). This may reflect the inclusion of late- and end-phase viral load measurements in the NFV trial, which were not available in the Illinois dataset. It should be noted that the primary aim of applying mathematical modeling in this study was not to dissect viral mechanisms in detail, but to reconstruct viral dynamics for stratification purposes. Accordingly, we adopted the simpler model [i.e., Eqs.(1-2)], which yields comparable dynamics while being more practical and sufficient for reconstructing viral load in this context. This approach also avoids the additional complexity and parameter assumptions required by Eqs.(3-6), as described in ⁵.

Next, to stratify the time-course pattern of viral shedding, we first applied unsupervised random forest clustering to the individual “reconstructed” virus dynamics of 144 participants (e.g., **Supplementary Fig 1**). However, this analysis failed to divide the time-course pattern into different clusters (data not shown). To overcome this problem, we quantified the peak, duration, up-slope (i.e., growth rate), and down-slope (i.e., decay rate) of the reconstructed dynamics as “features” of the virus dynamics (**Supplementary Table 4**). Interestingly, the unsupervised random forest clustering based on these features identified 3 groups (i.e., G1: N=46, G2: N=61, and G3: N=37) in which the time-course patterns were clearly discriminated. This finding suggested that there is a heterogeneity of virus infection dynamics in saliva (see Methods). **Fig 2A** indicates a two-dimensional Uniform Manifold Approximation and Projection (UMAP) embedding of the three stratified groups. Using a different color for each group (gray for G1, magenta for G2, and blue for G3), we also plotted the estimated individual viral load (**Fig 2B**), and the highlighted time-course pattern of each group by the Partial Least-Squares Discriminant Analysis (PLS-DA) (**Fig 2C**). In addition, we evaluated the impact of statistical uncertainty in the estimated viral dynamics on the stratification by comparing the uncertainty-adjusted and original distance matrices using the Mantel test (see Methods). The two matrices were strongly correlated (Mantel $r = 0.72$, $p < 0.001$), indicating that the stratification is robust to parameter uncertainty.

Clinical data	Unit	Group1 (N=33)	Group2 (N=37)	Group3 (N=20)	Overall (N=90)
Basic demographic information					
Age	years	47 (13.9)*	40.1 (12.6)	43.4 (13.7)	43.7 (13.6)
Sex***	-	46%	67%	70%	59%
Vital signs					
Systolic blood pressure	mmHg	116 (16.3)	119.8 (14.4)	122.2 (20.3)	118.8 (16.6)
Diastolic blood pressure	mmHg	77.5 (14.3)	75.9 (10.9)	78.5 (14.8)	77.1 (13.2)
Pulse rate	bpm	83.5 (16.9)	86.9 (14.3)	79.1 (13.1)	83.8 (15.3)
SpO ₂	%	97.5 (1)	97.7 (0.8)	97.6 (0.9)	97.6 (0.9)
Respiratory rate	bpm	17 (3.9)	16.5 (3.4)	16.8 (2.4)	16.8 (3.4)
Blood test results					
White blood cell count	10 ⁹ /L	5.1 [30.2]**	5.3 [31.9]	5.9 [34]	5.3 [32.8]
Neutrophil	%	62.4 (10.3)	60 (11.1)	59.8 (11.4)	61 (10.8)
Eosinophil	%	1 (1.8)	0.9 (1.5)	1 (1.2)	1 (1.6)
Basophil	%	0.3 (0.2)	0.4 (0.3)	0.4 (0.3)	0.3 (0.3)
Lymphocytes	%	26.1 (8.4)	28.1 (10)	28 (9.2)	27.3 (9.1)
Monocyte	%	10.1 (3.7)	10.6 (3.5)	10.8 (4.2)	10.4 (3.7)
Red blood cell count	10 ¹² /L	5 [450.3]	5.5 [464.1]	5.4 [450.9]	5.3 [453]
Amount of hemoglobin	g/dL	14 (1.4)	15.1 (1.1)	14.8 (1.3)	14.6 (1.4)
Hematocrit	%	41.6 (3.6)	44.1 (3.2)	43.8 (3.6)	43 (3.6)
Platelet count	10 ⁹ /L	121.6 (88.7)	127.8 (98)	136.6 (102.4)	127.2 (94.4)
CRP	mg/dL	1.3 (2.6)	1.3 (1.9)	1.1 (1.8)	1.3 (2.2)
Protein	g/dL	7.4 (0.4)	7.4 (0.4)	7.4 (0.3)	7.4 (0.4)
Albumin	g/dL	4.2 (0.3)	4.3 (0.3)	4.4 (0.3)	4.3 (0.3)
ALT	U/L	20.2 (11.9)	29.9 (23.9)	31.3 (22)	26.2 (19.8)
AST	U/L	21.9 (7.8)	28.3 (14)	28.7 (13.4)	25.8 (12)
γ-GTP	U/L	30.9 (25.6)	42 (41.8)	47.8 (35)	38.7 (34.7)
ALP	U/L	69.9 (20.9)	72 (20.3)	72.8 (20.6)	71.3 (20.4)
LDH	U/L	172.1 (34.1)	184.1 (43.8)	186.6 (29.5)	179.7 (37.2)
Bilirubin	mg/dL	0.5 (0.2)	0.5 (0.2)	0.6 (0.2)	0.5 (0.2)
CK	U/L	85.2 (48.2)	126.7 (146.5)	118.2 (133)	107.7 (113.1)
Na	mEq/L	139.3 (2.9)	139.9 (2.2)	139.8 (2.8)	139.6 (2.6)
K	mEq/L	3.9 (0.3)	3.9 (0.3)	3.9 (0.2)	3.9 (0.3)
Cl	mEq/L	102.6 (3.3)	102.7 (3.6)	102.8 (2.7)	102.6 (3.3)
BUN	mg/dL	11.4 (3)	11.3 (2.5)	12.5 (3.1)	11.6 (2.9)
CRE	mg/dL	0.8 (0.2)	0.8 (0.2)	0.9 (0.2)	0.8 (0.2)
Blood glucose	mg/dL	105.9 (19.9)	102.4 (17.1)	109 (15.7)	105.3 (18)
HbA1c	%	5.6 (0.3)	5.5 (0.4)	5.6 (0.3)	5.6 (0.3)
Procalcitonin	μg/L	0.1 (0.2)	0.1 (0.1)	0.1 (0.04)	0.1 (0.1)
Fibrinogen	mg/dL	371 (94.8)	365.3 (101.1)	349.2 (73)	364.1 (92.3)
PT	%	103.1 (20.5)	105.5 (17.5)	109.3 (15.5)	105.4 (18.4)
APTT	sec	31.4 (3.7)	31.1 (5.3)	30.7 (3.2)	31.1 (4.2)
PT-INR	-	1 (0.1)	1 (0.1)	1 (0.1)	1 (0.1)
D-dimer	mg/L	0.8 (1.3)	0.5 (0.2)	0.5 (0.1)	0.6 (0.8)

* Mean (standard deviation) and **median [Interquartile range]. ***Sex was reported as the proportion of males.

Table 1.

Clinical data of the overall cohort from the NFV clinical trial and in groups

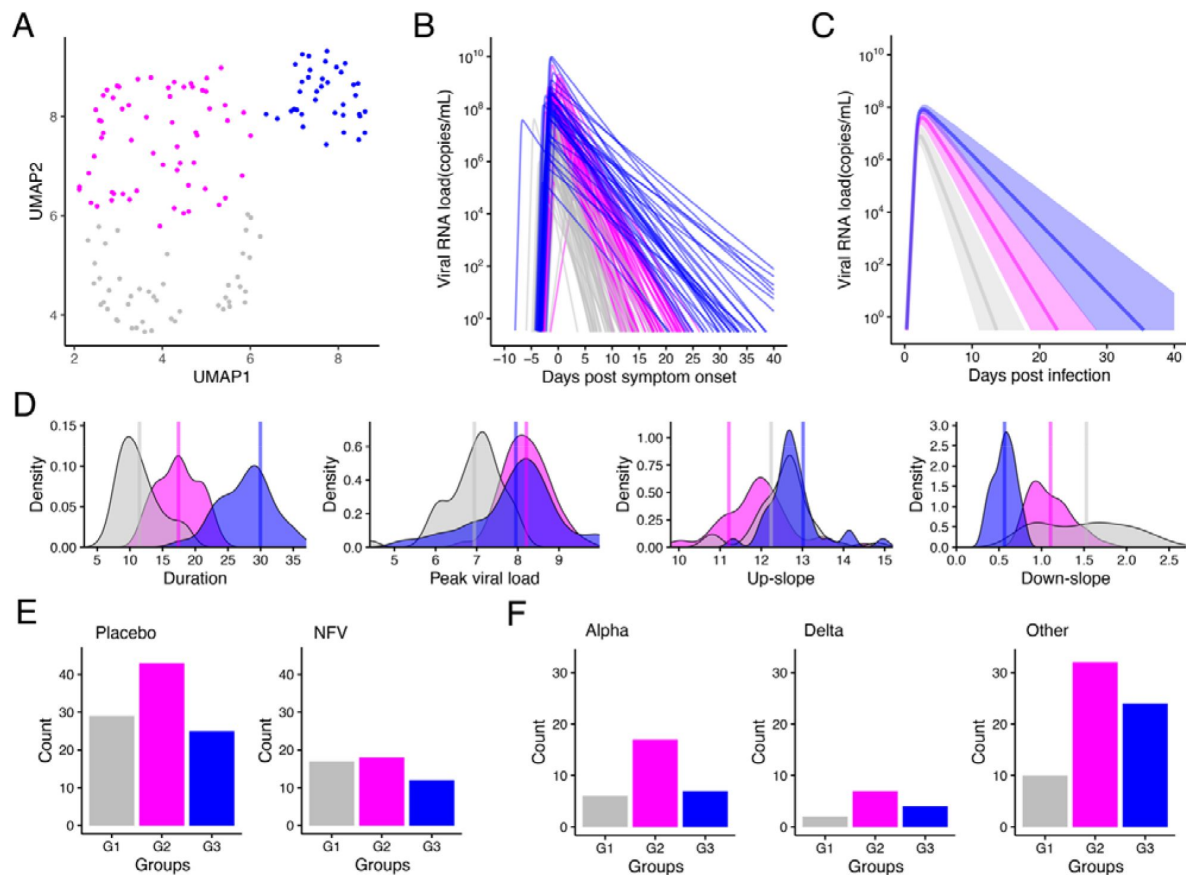


Figure 2.

Stratification of individual SARS-CoV-2 viral dynamics in saliva:

(A) UMAP of stratified viral RNA load based on the extracted features from the reconstructed individual-level viral dynamics is shown. (B) The reconstructed individual viral RNA load is shown. Colors for individual-level viral dynamics correspond to the colors of the dots in the UMAP described in (A). (C) The time-course patterns of each group highlighted by the Partial Least Squares Discriminant Analysis (PLS-DA). (D) Distributions between groups of each feature used for stratification of viral shedding patterns are shown. The p-values of ANOVA for the difference in each feature among stratified group are all less than 0.05. (E) Distributions of the number of individuals in each stratified group for the standard-of-care alone (left, n=97) and standard-of-care plus NFV administration (right, n=47) participants are shown. (F) Distributions of the number of individuals in each stratified group for Alpha variants (left, n=30), Delta variants (middle, n=13), and other variants (right, n=66) of SARS-CoV-2 are shown.

The distributions of the four features are described in [Fig 2D](#); a statistically significant between-group difference was found in the duration of viral RNA detection. The mean durations were 11.5 days (95% CI: 10.6 to 12.4), 17.4 days (16.6 to 18.2), and 30.0 days (28.1 to 31.8) for G1, G2, and G3, respectively. In our previous report¹⁰, we consistently confirmed that there were at least 3 groups showing different duration of viral RNA detection in upper respiratory specimens.

Because of previous work concluding that there are no significant differences in virus infection dynamics between NFV-treated and untreated participants¹², we analyzed all data together regardless of treatment (see Methods). To further confirm whether NFV affects the stratification of the time-course pattern of viral shedding, we compared the number of individuals belonging to each group (i.e., G1, G2 and G3) between NFV-treated and untreated participants (including the members of the University of Illinois cohort) and found no trend for the stratification ($p=0.784$ by the Fisher's exact test: [Fig 2E](#)).

Another possibility that explains the different viral duration observed here may be a difference in VOC genotypes. To test this, we used data from 55 participants in our NFV clinical trial who had been characterized according to which VOCs [i.e., B.1.1.7 (Alpha), B.1.672.2 or AY.29 (Delta), and other variants] they had been infected with (see Methods), in addition to data from 54 participants of the University of Illinois cohort. However, we observed no trend in the number of individuals belonging to each group among the VOCs ($p=0.728$ by the Fisher's exact test: [Fig 2F](#)), which is consistent with the conclusion in Ke et al.⁵

Basic clinical data may not explain heterogeneity in individual viral shedding

Using data from the NFV clinical trial, we annotated the saliva viral loads of 90 participants with basic demographic information, daily symptoms, blood test results, and vital signs ([Fig 1A](#), [Table 1](#) and [Supplementary Table 1](#)). We also annotated the saliva viral loads of 52 participants from the University of Illinois with daily symptoms ([Supplementary Table 1](#)).

To identify factors that were significantly correlated with the viral shedding patterns in saliva specimens obtained from the NFV clinical trial, we first examined the 39 variables summarized in [Table 1](#). Each factor was compared between the three groups by ANOVA and the p-values were corrected by the False Discovery Rate (FDR). However, we found no clinical data that differed significantly (i.e., corrected p-value of ANOVA of less than 0.05) among the stratified groups ([Fig 3A](#)). To avoid overfitting by bootstrap aggregating (bagging), we also trained a random forest classifier (see Methods), a tree-based machine learning algorithm suitable for tabular data¹⁴, to predict the group from the clinical data of 90 individuals in the NFV clinical trial cohort and obtained ROC-AUCs of 60%, 49%, and 36% for predicting G1, G2, and G3, respectively ([Fig 3B](#)). We were not able to achieve a high ROC-AUC for predicting the shedding patterns based on the basic clinical data. We also attempted to make prediction based on clinical data which exhibited relatively low p-values in the ANOVA analysis. However, we were unable to achieve a high prediction accuracy with this approach (data not shown).

Next, we asked whether the stratification of the study population is associated with clinical symptoms of COVID-19 that could be caused by active replication of SARS-CoV-2. In general, the clinical symptoms of COVID-19 are cough, fever, shortness of breath, muscle pain, sore throat, confusion, chest pain, headache, rhinorrhea, diarrhea, and nausea and vomiting. In our study, individual-level symptom data were available as 8 categories in the cohorts from both the NFV clinical trial and the University of Illinois ([Supplementary Table 1](#)). Symptom data were obtained via participant self-report and encoded as categorical indicators (presence/absence) for each symptom. We tried to use a random forest classifier to investigate whether symptom data could predict the stratified groups and obtained ROC-AUCs of 57%, 54%, and 48% for predicting G1,

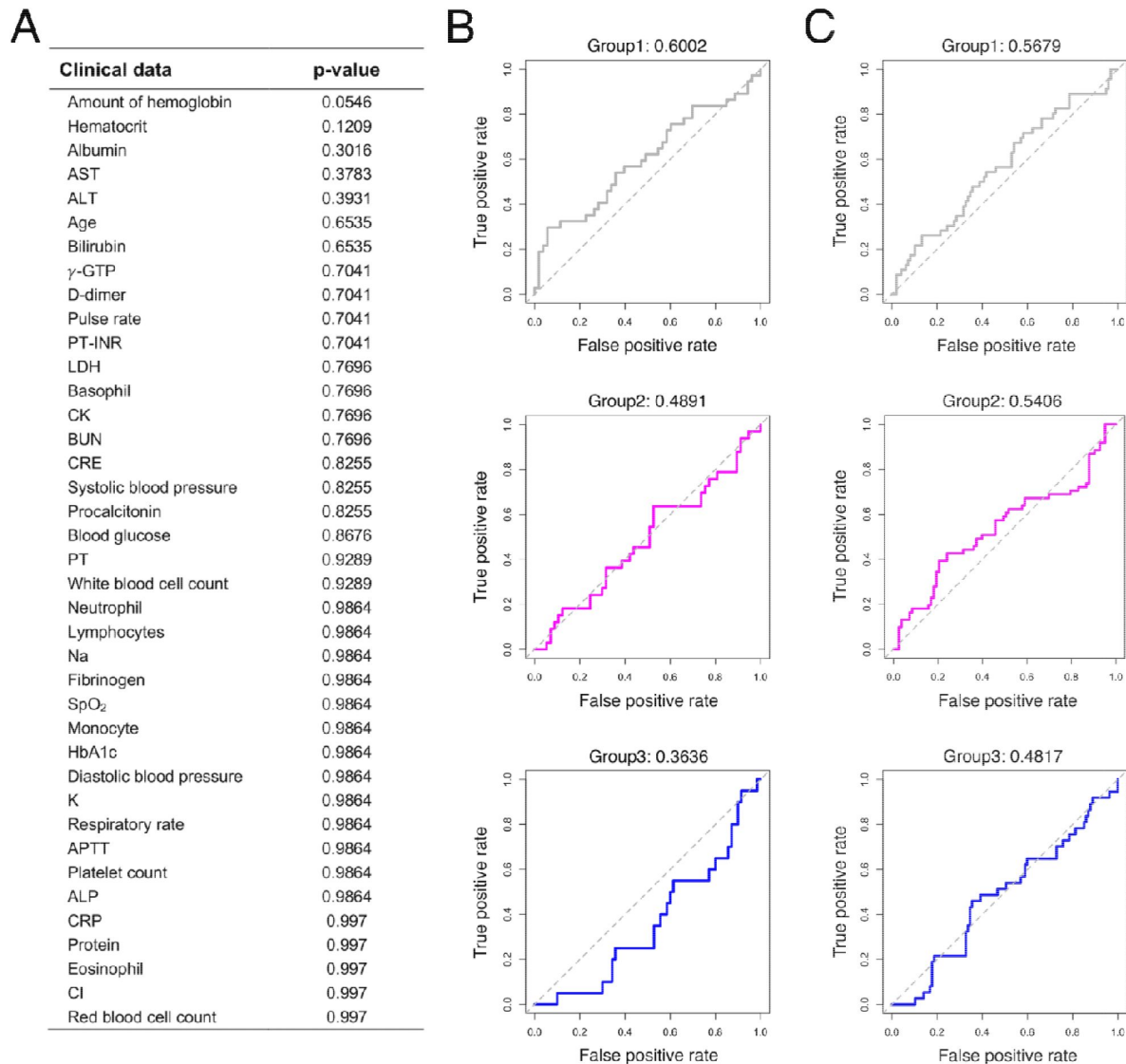


Figure 3.

Correlation between clinical data and viral shedding patterns:

(A) P-values of ANOVA corrected by the FDR to compare clinical data among the three stratified groups are shown. Clinical data are listed in reverse order of p-values. **(B)** and **(C)** show ROC curves of random forest classifiers trained on predicting each group by using data for **(B)** clinical values described in [Table 1](#) and **(C)** symptom onset data described in [Supplementary Table 1](#), respectively. The corresponding AUC (area under curve) value of each ROC curve is shown on the top of each panel.

G2, and G3, respectively (**Fig 3C**). In fact, SARS-CoV-2 human challenge clearly showed no quantitative correlation between the individuals' time-series pattern of viral load and symptoms¹⁵.

Additionally, we investigated the relationship between each feature of viral dynamics (i.e., duration of viral RNA detection, peak viral load, up-slope, and down-slope) and the clinical data by using the Pearson's correlation coefficient (Supplementary Table 5). Overall, correlation coefficients were low (0.06 on average) with high p-values, which suggests that no feature was likely to be explained by these clinical data.

Relationship between salivary micro-RNAs and viral shedding patterns in COVID-19 patients

Various proteins in saliva have antiviral effects. It is also expected that some micro-RNAs in saliva may impair SARS-CoV-2 replication¹. In addition, saliva-based micro-RNAs have already been reported to be associated with various diseases^{16–21}, including COVID-19²². Based on this, we anticipated that among the biomarkers obtainable from saliva, micro-RNAs could potentially serve as non-invasive predictors of viral dynamics. Accordingly, despite being less readily obtainable than clinical information, micro-RNAs are accessible from saliva sample and were anticipated to be closely linked to viral dynamics, so we incorporated them into our analysis.

We here used the stored residual saliva specimens from the NFV clinical trial to identify micro-RNA(s) associated with the stratified groups (i.e., G1, G2 and G3). We note that, because all residual saliva specimens are annotated with the individual participant and we know which participants belong to which stratified group, we can select and compare saliva specimens from G1, G2, and G3 without bias. This implies that we can impartially select participants in equal numbers from each group, unaffected by other factors. Specifically, we collected 60 stored saliva specimens from the NFV clinical trial to perform micro-RNA analysis for 30 participants. We picked two samples for each participant to evaluate the role of micro-RNAs during both the peak and the late phase (i.e., 30 samples for each phase): the nearest sample from the estimated peak and the most distant sample above the detection limit in the late phase (**Fig 4A**). We normalized micro-RNA expression among the samples by using the DESeq2 Bioconductor package. We summarize the information on the micro-RNAs we obtained from saliva specimens in **Supplementary Table 6**.

First, we compared micro-RNA levels between two phases using pairwise t-tests (and Mann-Whitney U tests) with FDR correction. As a result, no micro-RNA showed a statistically significant difference. This suggests that micro-RNA levels remain relatively stable during the course of infection and may therefore serve as a biomarker independent of sampling time.

Next, similar to the analysis using clinical data, we compared the expression levels of 92 micro-RNAs between the three stratified groups. Because the micro-RNA data were non-parametric, we used the Kruskal-Wallis ANOVA for analysis and corrected the p-values by FDR. However, we failed to find micro-RNA levels that differentiated the stratified groups (i.e., with a corrected p-value of Kruskal-Wallis ANOVA of less than 0.05) in the three trials using the data from the peak phase, the late phase, and both phases (e.g., **Fig 3B** for the total 60 samples). We also trained a random forest classifier to predict each group from the micro-RNA levels for the 60 total samples, and obtained ROC-AUCs of 48%, 57%, and 42%, respectively. Again, we did not obtain enough ROC-AUCs to predict stratified groups by using the collected micro-RNA data.

Furthermore, we investigated the relationship between the 4 features of viral dynamics and micro-RNA levels. Here we used the Spearman's correlation coefficient (**Supplementary Table 7**). Overall, we did not find strong correlations between micro-RNA levels and features (Spearman's correlation coefficients on average of 0.002, 0.024, -0.001, and -0.001 for duration of viral RNA detection, peak viral load, up-slope, and down-slope, respectively, for the 60 total

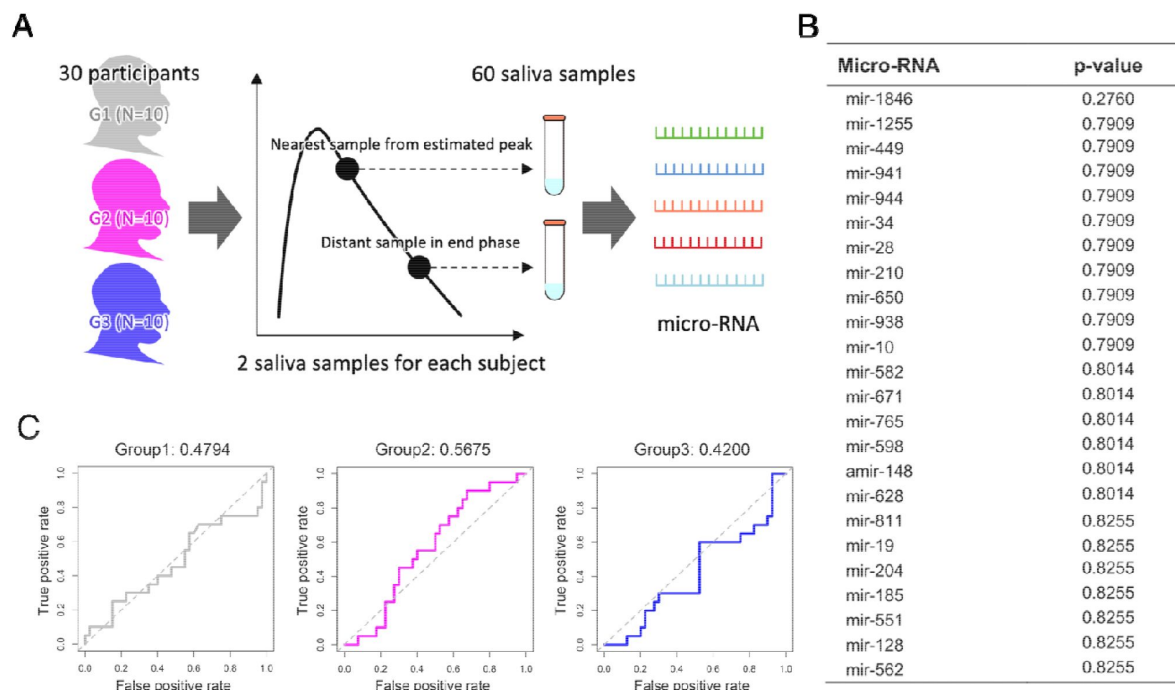


Figure 4.

Correlation between micro-RNA data and viral shedding patterns:

(A) The strategy of micro-RNA data collection from saliva samples in the NFV clinical trial is described. We picked a total of 30 participants by choosing 10 participants from each group. We chose two samples (the nearest to estimated peak and the most distant but above the detection limit in the late phase) from each participant for quantifying micro-RNA. **(B)** P-values of Kruskal-Wallis ANOVA corrected by the FDR for each micro-RNA level are shown. Micro-RNA levels are listed by reverse order of p-values. Only the 24 micro-RNA levels with the lowest p-values are shown. **(C)** ROC curves of random forest classifiers trained on predicting each group by using levels of 92 micro-RNAs are shown. The corresponding AUC value of each ROC curve is presented on the top of each panel.

samples). Only the mir-1846 level exhibited a weak negative correlation (-0.53 Spearman's correlation coefficient with 0.01 p-value) with the peak viral load (**Supplementary Fig 5**). We confirmed similar trends even when we analyzed the micro-RNA level for the peak and late phases separately.

Discussion

Being able to quickly and efficiently diagnose COVID-19 is essential in monitoring the pandemic. Because the sampling process for saliva is noninvasive, and because it is inexpensive and minimizes the risk for transmissions to health care workers¹, saliva sampling has excellent potential and advantages over other sampling methods from biological specimens such as the lower and upper respiratory tract^{2,23}. Given the significant individual heterogeneity in the saliva viral shedding^{5,24}, identifying biomarker(s) for viral shedding patterns will be crucial for improving public health interventions in the era of living with COVID-19.

To improve our understanding of SARS-CoV-2 infection dynamics in saliva to enable application of saliva testing in the fight against COVID-19, we quantified and stratified longitudinal virus dynamics in saliva samples from 144 mildly symptomatic individuals from the cohorts of the NfV clinical trial¹¹ and the University of Illinois at Urbana-Champaign⁵, and we uniquely analyzed the relationships among viral dynamics, clinical data, and micro-RNAs. Our mathematical modeling analysis indicates that viral dynamics in saliva may exhibit distinct patterns compared to those in the upper respiratory tract. We estimated that viral replication in saliva is characterized by a relatively rapid early growth phase, with a mean (standard deviation) doubling time of 1.44 (0.49) hours. Compared with prior studies analyzing viral dynamics in the upper respiratory tract using similar models, which reported doubling times of 2-4 hours^{5,25,26}, our findings suggest that viral replication in saliva proceeds faster than in the upper respiratory tract. Multiple previous studies have also shown that viral loads in saliva rise more rapidly than in the nasal cavity, are detected with higher sensitivity early in infection, and reach their peak earlier^{5,27,30}.

In addition to the large heterogeneity in virus infection dynamics, we identified three groups (i.e., G1, G2 and G3) with different viral shedding patterns (**Fig 2D**). Immunocompromised patients have been reported to have a prolonged duration of viral RNA detection, lasting over three months, underscoring the critical role of host immune responses in controlling viral infections³¹⁻³⁴. Although oral immune responses remain poorly understood, Huang et al. recently confirmed by using single-cell RNA sequencing of the human minor salivary glands and gingiva that SARS-CoV-2 infection can trigger sustained, localized immune responses in saliva³. In this study, we observed significant differences in the down-slopes of viral shedding in saliva among participants in different groups, with a more rapid decline in G1. This decline is likely attributed to a stronger immune response to SARS-CoV-2 in G1 participants than in participants in G2 and G3, as reflected in the death rate of infected cells due to the immune response (**Fig 2D**). Lower levels of viral replication have also been observed among infected participants with high baseline levels of mucosal IgA (but not IgG), as reported elsewhere³⁵. Recently, we demonstrated that rapid anti-spike secretory IgA antibody responses can contribute to reducing duration of viral RNA detection and amounts in nasopharyngeal mucosa³⁶. These findings highlight the importance of biomarkers that directly reflect an individual's immune response, such as virus-specific antibody induction and T cell levels etc., in predicting viral shedding patterns. Therefore, quantifying the time-series pattern of mucosal IgA and its correlation with saliva viral load may provide crucial insights into the stratification of SARS-CoV-2 infection dynamics.

For the purpose of predicting viral shedding patterns during the early stage of infection, we first explored the association of 39 basic clinical variables, 8 daily symptoms, and the levels of 92 micro-RNAs with the stratified groups. However, none of the factors were significant (**Table 1**,

Fig 3A, **Fig 4B**, **Supplementary Table 1** and **Supplementary Table**). On the other hand, it is noteworthy that all infection cases were mild and that most participants had clinical indicators within normal ranges. This lack of clinical heterogeneity within the cohort may have limited the ability to fully capture the diversity of infection dynamics. Moreover, the clinical parameters analyzed in this study are, a priori, unlikely to exhibit strong correlations with virologic outcomes. In contrast, we showed that mir-1846, which is an exogenous micro-RNA that is specifically classified as an *Oryza sativa* micro-RNA (osa-microRNA)³⁷, may exhibit a weak negative correlation. Exogenous micro-RNAs enter the human body primarily through food and can affect human metabolism by interacting and binding with human genes. mir-1846 is reported to interact with two human genes³⁷ that are known to be associated with the progression of melanoma, various cancers, and leukemia. This suggests that mir-1846 levels may be linked to human immunity. Few studies have investigated the role of mir-1846 in humans, but our findings suggest the need for further investigations into the impact of this micro-RNA level on human immunity. Our research sheds light on the intricate patterns of viral shedding in saliva.

Our approach has several limitations that must be considered in our next study: First, our analysis was limited to participants with symptomatic infection and excluded those with asymptomatic infection (22 asymptomatic individuals out of a total of 182 individuals, i.e., 12% of participants) because we integrated datasets with different time scales from different cohorts. Although our data do not include participants infected with omicron variants, others have reported that the omicron variant may cause a higher proportion of asymptomatic infection³⁸. Thus, evaluating the effect of asymptomatic infection will be important to update our stratification, especially for recent (or future emerging) VOCs. Second, potential viral rebound was neither prespecified nor systematically assessed. A subset of participants exhibited patterns consistent with possible rebound (e.g., S01-16 and S01-43 in **Supplementary Fig 1**), which could affect estimates of viral RNA detection duration. Future studies should predefine an operational criterion for viral rebound and explicitly incorporate it into the modeling framework to strengthen robustness. Third, micro-RNAs participate in the post-transcriptional regulation of gene expression; however, they do not provide direct insights into immune cell dynamics. Given the reported association between the duration of viral RNA detection and mucosal immunity as discussed above, it appears imperative to analyze modalities that are directly linked to the immune response in the future. Fourth, some of our results may have limited relevance to the current COVID-19 situation, as most people have now either been infected or vaccinated. Nevertheless, investigating the relationship between viral shedding patterns in saliva and various clinical and microRNA data, and developing a method to do so, remains important. Such research can offer valuable insights into the early response to emerging infectious viruses in the future.

Another potential limitation of this study is the timing of saliva specimen sampling, although we took great care to select and compare specimens from G1, G2, and G3 without bias. As a result of our clinical trial design (jRCT2071200023^{11,12}), participants were enrolled after the onset of symptoms, thereby restricting saliva specimen collection exclusively to the post-symptom phase. Unfortunately, we lack samples from the pre-infection, pre-symptomatic, and early infection phases. Consequently, the absence of individual-level baseline values for micro-RNA means that inter-participant heterogeneity in micro-RNA levels may obscure signals related to distinct viral infection dynamics in saliva.

In conclusion, our study revealed that the dynamics of SARS-CoV-2 infection in saliva can be classified into three groups based mainly on the duration of viral RNA detection. However, accurately predicting the variability in viral dynamics remains a challenging task, because it requires a more comprehensive understanding of the complex shedding patterns in saliva, as well as detailed clinical and molecular data. The identification of a sensitive, simple, and rapid biomarker for saliva viral shedding will be imperative for future COVID-19 outbreak control.

Methods

Ethics statement

The NFV clinical trial was approved by the institutional review board of Nagasaki University Hospital (approval number: I20-001) and is registered with the Japan Registry of Clinical Trials (jRCT2071200023). All participants provided written, informed consent for secondary use of clinical information and samples. The present study was approved by the ethics committee of Nagoya University (approval number: hc 22-01).

Saliva viral load data

Longitudinal saliva viral load data of participants with symptomatic and asymptomatic COVID-19 (122 cases) were obtained from the NFV clinical trial¹¹. Briefly, the NFV clinical trial was a prospective, randomized, open-label, blinded-endpoint, parallel-group trial conducted between July 2020 and October 2021 at 11 university and teaching hospitals in Japan. This study consisted of a 14-day treatment period and a 14-day follow-up period, with no significant differences in the time to viral clearance between patients who received standard-of-care plus NFV administration and those who had the standard-of-care alone^{11,12}. Therefore, the participants with COVID-19 were analyzed together here. In addition, we obtained similar saliva viral load data (60 cases) from the cohort of the University of Illinois at Urbana-Champaign⁵. This cohort contained all faculty, staff, and students at the University of Illinois at Urbana-Champaign, who undergo at least twice weekly quantitative PCR-RT testing during fall of 2020 and spring of 2021. The viral load data for Illinois cohort was calculated based on the linear relationship between viral load (copies/mL) and Ct values presented in⁵, specific to the measurement method used in this study. Among those 182 cases, we focused only on symptomatic participants. Asymptomatic individuals were excluded since they lack a definable date of symptom onset, which prevents us from establishing a clear baseline for the time axis in parameter estimation. Also, we excluded the participants who had less than three measured viral loads that were not limit detections (i.e., 90 cases from the NFV clinical trial and 54 cases from the University of Illinois were used in this study). Because the trial design discontinued follow-up after two consecutive results below the detection limit, most of these participants already had viral loads near the detection limit at baseline, making reliable model fitting infeasible. The limit of detection for viral load data from the University of Illinois is 1.08 copies/ml. However, the limit of detection for viral load data from NFV clinical trial was unclear. Considering this, we assumed 1.08 copies/ml as the limit of detection for all viral load data. To examine the impact of assumptions regarding the detection limit, we performed a sensitivity analysis. Specifically, for the NFV cohort, we compared the estimated viral dynamics when the detection limit was set to 0 and 2, instead of 1.08. Overall, the results were largely consistent across these scenarios (Supplementary Fig 6).

Viral genome sequencing

The cDNA had been synthesized from RNA of SARS-CoV-2-positive saliva samples. Reverse transcription, multiplex PCR reaction, and Illumina library prep were conducted using a protocol published previously³⁹. The pooled library was first purified by AMPureXP at 0.8 x concentration and then again at 1.2 x concentration. The purified library was sequenced for 151 cycles at both paired-ends in Illumina iSeq100. Sequence analysis was performed using the nf-core/viralcon pipeline (10.5281/zenodo.3901628).

Quantifying biomarkers in saliva

Total RNA from saliva was extracted with MagMAX CORE Nucleic Acid Purification kits (Applied Biosystems, Foster City, CA). Micro-RNAs were detected using Illumina HiSeq x Ten (Illumina, Inc, San Diego, CA) with data processing by ribodepletion (Genewiz-Azenta, South Plainfield, NJ). To remove technical sequences, the pass filter data in the fastq format were processed by Trimmomatic (v0.30) to be high-quality clean data. Following quality trimming, micro-RNAs were identified and checked using miRDeep2⁴⁰. Normalization of micro-RNA expression among samples and differential expression analysis were carried out using the DESeq2 Bioconductor package.

Basic clinical data

Basic clinical data including basic demographic characteristics of the study participants, symptoms, and findings of physical examinations and laboratory tests were obtained according to the study protocol¹¹. We here used information on age, daily symptoms, results of blood tests, and vital signs of the symptomatic participants in the NFV clinical trial (summarized in **Table 1** and **Supplementary Table 1**).

Mathematical modeling

To describe SARS-CoV-2 infection dynamics in saliva specimens, we here mainly used the following mathematical model developed in our recent studies^{4,6,10}:

$$\frac{df(t)}{dt} = -\beta f(t)V(t), \quad (1)$$

$$\frac{dV(t)}{dt} = \gamma f(t)V(t) - \delta V(t). \quad (2)$$

The variables $f(t)$ and $V(t)$ are the fraction of uninfected target cells and the total amount of virus, respectively, and the parameters β , γ , and δ are the rate constant for virus infection, the maximum rate constant for viral replication, and the death rate of infected cells, respectively.

In addition to comparing the simple model [i.e., a target-cell-limited model; Eqs.(1-2)], we also used the following “immune effector cell model” developed in Ke et al.⁵ for the saliva viral load (see **Supplementary Fig 4**):

$$\frac{dT(t)}{dt} = -\beta T(t)V(t), \quad (3)$$

$$\frac{dE(t)}{dt} = \beta T(t)V(t) - kE(t), \quad (4)$$

$$\frac{dI(t)}{dt} = kE(t) - \delta(t)I(t), \quad (5)$$

$$\frac{dV(t)}{dt} = \pi I(t) - cV(t). \quad (6)$$

The variables $T(t)$, $E(t)$, and $I(t)$ are the total number uninfected target cells, cells in the eclipse phase of infection, and productively infected cells, respectively. The parameters $1/k$, π , and c are the average duration of the eclipse phase, the virus production rate, and the clearance rate of viruses, respectively. The death rate of infected cells is assumed to be time-dependent to mimic the killing of infected cells by immune effector cells: $\delta(t) = \delta_1$ for $t < t_1$ and $\delta(t) = \delta_1 + \delta_2$ for $t_1 < t$. For a detailed explanation of the immune effector cell model, the reader is referred to Ke et al.⁵.

Parameter estimation

A nonlinear mixed-effects modelling approach incorporates fixed effects as well as random effects that describe the inter-participant variability in parameters. Including random effects amounts to a partial pooling of the data of all participants to improve estimates of the parameters applicable across the cases. In this study, viral load data from the Illinois cohort were collected primarily during the early to middle phase of SARS-CoV-2 infection, whereas those from the NFV cohort were obtained mainly during the middle to late phase. In particular, the Illinois cohort provides pre-symptomatic data, while the NFV cohort includes data beyond 14 days after symptom onset, with each cohort lacking the phase represented in the other. Although the NFV cohort lacks pre-symptomatic data, the nonlinear mixed-effects model first derives population-level parameters from all participants and then accounts for individual variability through random effects. This allows pre-symptomatic information from the Illinois cohort to inform the inferred viral dynamics of participants in NFV cohort. Conversely, for the late phase, the NFV cohort serves the complementary role. By jointly analyzing both cohorts with a nonlinear mixed-effects model to estimate individual-level parameters, we can therefore capture the complete time-course pattern of SARS-CoV-2 infection dynamics across all participants, to the extent possible.

In our analyses, the variable $V(t)$ in Eq.(2) corresponds to the viral load in saliva for SARS-CoV-2. To fit the patient's viral load data, we used a program MONOLIX 2021R2 (www.lixoft.com), implement maximum likelihood estimation of parameters in nonlinear mixed effect model. The nonlinear mixed effect model allows a fixed effect as well as interpatient variability. This method estimates each parameter $\theta_i (= \theta \times e^{\eta_i})$ for each individual where θ is a fixed effect, and η_i is a random effect, and which obeys a Gaussian distribution with mean 0 and standard deviation Ω . Here we used lognormal distributions as prior distributions of parameters to guarantee the positiveness (i.e., negative values do not biologically make sense). In parameter estimation, as time 0 in the original dataset represents the time of symptom onset, we also estimated the time from infection to symptom onset (corresponding to τ in *Supplementary Table 2* and *Supplementary Table 3*) along with other parameters. The fixed effect parameters and random effect parameters were estimated by use of the stochastic approximation Expectation/Maximization (SAEM) algorithm and empirical Bayes method, respectively. A right-truncated normal distribution was used in the likelihood function to account for the left censoring of the viral load data (i.e., when the viral load is not detectable)⁴¹. The standard errors were obtained from the Fisher Information Matrix using the linearization method (*Supplementary Fig 3C*). We changed the initial values multiple times to avoid a local minimum of AIC and confirmed the robustness of parameter estimation.

Unsupervised clustering and stratification of SARS-CoV-2 infection dynamics

Unsupervised random forest clustering was performed on the selected features of the virus infection dynamics, that is, the peak viral load, duration of viral RNA detection, up-slope, and down-slope (rfUtilities package in R). The use of random forest allows us to avoid overfitting by bootstrap aggregating (bagging) and to achieve better generalization performance⁴². After a random forest dissimilarity (i.e., the distance matrix between all pairs of samples) was obtained, it was visualized with Uniform Manifold Approximation and Projection (UMAP) in a two-dimensional plane and was stratified with spectral clustering (Python scikit-learn). The optimal number of clusters was determined by the eigengap heuristic method.

Furthermore, to evaluate how statistical uncertainty in the estimated viral dynamics might influence the stratification, we propagated parameter uncertainty as follows. For each participant, we resampled the parameters of the mathematical model 100 times within their 95% credible intervals and calculated the corresponding features of reconstructed viral dynamics. We then averaged the features across the 100 replicates to obtain an uncertainty-adjusted feature vector

for each participant. Using these vectors, we re-calculated the distance matrix as described above. Then agreement between the uncertainty-adjusted and original distance matrix was quantified with a permutation-based Mantel test.

Random forest classifiers for characterizing stratified groups

Random forest classifiers were trained to predict either of the three stratified groups (G1-G3) using randomForest packages in R. The receiver operating characteristic (ROC) curve for each classifier was drawn from out-of-bag (OOB) samples using the pROC package in R. For example, the ROC for G1 is for the random forest classifier predicting “G1” versus “G2 or G3”. We list the variables for the supervised random forest in [Table 1](#), [Supplementary Table 1](#), and [Supplementary Table 2](#).

Statistical analysis

When necessary, the variables were compared among different groups using Fisher’s exact test (for categorical variables), analysis of variance (ANOVA, for numerical variables from clinical data with more than two groups), and Kruskal-Wallis ANOVA (for variables from micro-RNA data with more than two groups). Corrections of p-values for multiple testing were performed by the False Discovery Rate (FDR). Also, the variables were investigated for their relationship with features of viral load using the Pearson’s correlation coefficient (for variables from clinical data) and the Spearman’s correlation coefficient (for variables from micro-RNA data). All statistical analyses were performed using R (version 4.1.3).

Data availability

The datasets analyzed in this study are not publicly available owing to privacy and ethical considerations. They are, however, available from the corresponding author upon reasonable request, subject to ethics approval from all participating institutions.

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Additional files

Supplementary information [↗](#)

Additional information

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Reviewer #1 (Public Review):

Summary:

This study by Park and colleagues uses longitudinal saliva viral load data from two cohorts (one in the US and one in Japan from a clinical trial) in the pre-vaccine era to subset viral shedding kinetics and then use machine learning to attempt to identify clinical correlates of different shedding patterns. The stratification method identifies three separate shedding patterns discriminated by peak viral load, shedding duration, and clearance slope. The authors also assess micro-RNAs as potential biomarkers of severity but do not identify any clear relationships with viral kinetics.

Strengths:

The cohorts are well developed, the mathematical model appears to capture shedding kinetics fairly well, the clustering seems generally appropriate, and the machine learning analysis is a sensible, albeit exploratory approach. The micro-RNA analysis is interesting and novel.

<https://doi.org/10.7554/eLife.96032.2.sa3>

Reviewer #2 (Public Review):

Summary:

This study argues it has found that it has stratified viral kinetics for saliva specimens into three groups by the duration of "viral shedding"; the authors could not identify clinical data or microRNAs that correlate with these three groups.

Strengths:

The question of whether there is a stratification of viral kinetics is interesting.

<https://doi.org/10.7554/eLife.96032.2.sa2>

Reviewer #3 (Public Review):

The article presents a comprehensive study on the stratification of viral shedding patterns in saliva among COVID-19 patients. The authors analyze longitudinal viral load data from 144 mildly symptomatic patients using a mathematical model, identifying three distinct groups based on the duration of viral shedding. Despite analyzing a wide range of clinical data and micro-RNA expression levels, the study could not find significant predictors for the stratified shedding patterns, highlighting the complexity of SARS-CoV-2 dynamics in saliva. The research underscores the need for identifying biomarkers to improve public health

interventions and acknowledges several limitations, including the lack of consideration of recent variants, the sparsity of information before symptom onset, and the focus on symptomatic infections.

The manuscript is well-written, with the potential for enhanced clarity in explaining statistical methodologies. This work could inform public health strategies and diagnostic testing approaches.

Comments on the revised version from the editor:

The authors comprehensively addressed the concerns of all 3 reviewers. We are thankful for their considerable efforts to do so. Certain limitations remain unavoidable such as the lack of immunologic diversity among included study participants and lack of contemporaneous variants of concern.

One remaining issue is the continued use of the target cell limited model which is sufficient in most cases, but misses key datapoints in certain participants. In particular, viral rebound is poorly described by this model. Even if viral rebound does not place these cases in a unique cluster, it is well understood that viral rebound is of clinical significance.

In addition, the use of microRNAs as a potential biomarker is still not fully justified. In other words, are there specific microRNAs that have a pre-existing mechanistic basis for relating to higher or lower viral loads? As written it still feels like microRNA was included in the analysis simply because the data existed.

<https://doi.org/10.7554/eLife.96032.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review)

Summary:

This study by Park and colleagues uses longitudinal saliva viral load data from two cohorts (one in the US and one in Japan from a clinical trial) in the pre-vaccine era to subset viral shedding kinetics and then use machine learning to attempt to identify clinical correlates of different shedding patterns. The stratification method identifies three separate shedding patterns discriminated by peak viral load, shedding duration, and clearance slope. The authors also assess micro-RNAs as potential biomarkers of severity but do not identify any clear relationships with viral kinetics.

Strengths:

The cohorts are well developed, the mathematical model appears to capture shedding kinetics fairly well, the clustering seems generally appropriate, and the machine learning analysis is a sensible, albeit exploratory approach. The micro-RNA analysis is interesting and novel.

Weaknesses:

The conclusions of the paper are somewhat supported by the data but there are certain limitations that are notable and make the study's findings of only limited relevance to current COVID-19 epidemiology and clinical conditions.

We sincerely appreciate the reviewer's thoughtful and constructive comments, which have been invaluable in improving the quality of our study. We have carefully revised the manuscript to address all points raised.

(1) The study only included previously uninfected, unvaccinated individuals without the omicron variant. It has been well documented that vaccination and prior infection both predict shorter duration shedding. Therefore, the study results are no longer relevant to current COVID-19 conditions. This is not at all the authors' fault but rather a difficult reality of much retrospective COVID research.

Thank you for your comment. We agree with the review's comment that some of our results could not provide insight into the current COVID-19 conditions since most people have either already been infected with COVID-19 or have been vaccinated. We revised our manuscript to discuss this (page 22, lines 364-368). Nevertheless, we believe it is novel that we have extensively investigated the relationship between viral shedding patterns in saliva and a wide range of clinical and microRNA data, and that developing a method to do so remains important. This is important for providing insight into early responses to novel emerging viral diseases in the future. Therefore, we still believe that our findings are valuable.

(2) The target cell model, which appears to fit the data fairly well, has clear mechanistic limitations. Specifically, if such a high proportion of cells were to get infected, then the disease would be extremely severe in all cases. The authors could specify that this model was selected for ease of use and to allow clustering, rather than to provide mechanistic insight. It would be useful to list the AIC scores of this model when compared to the model by Ke.

Thank you for your feedback and suggestion regarding our mathematical model. As the reviewer pointed out, in this study, we adopted a simple model (target cell-limited model) to focus on reconstruction of viral dynamics and stratification of shedding patterns rather than exploring the mechanism of viral infection in detail. Nevertheless, we believe that the target cell-limited model provides reasonable reconstructed viral dynamics as it has been used in many previous studies. We revised manuscript to clarify this point (page 10, lines 139-144). Also, we revised our manuscript to provide more detailed description of the model comparison along with information about AIC (page 10, lines 130-135).

(3) Line 104: I don't follow why including both datasets would allow one model to work better than the other. This requires more explanation. I am also not convinced that non-linear mixed effects approaches can really be used to infer early model kinetics in individuals from one cohort by using late viral load kinetics in another (and vice versa). The approach seems better for making populationlevel estimates when there is such a high amount of missing data.

Thank you for your feedback. We recognized that our explanation was insufficient by your comment. We intended to describe that, rather than comparing performance of the two models, data fitting can be performed with same level for both models by including both datasets. We revised the manuscript to clarify this point (page 10, lines 135-139).

Additionally, we agree that nonlinear mixed effects models are a useful approach for performing population-level estimates of missing data. On the other hand, in addition, the nonlinear mixed effects model has the advantage of making the reasonable parameter estimation for each individual with not enough data points by considering the distribution of parameters of other individuals. Paying attention to these advantages, we adopted a nonlinear mixed effects model in our study. We also revised the manuscript to clarify this (page 27, lines 472-483).

(4) Along these lines, the three clusters appear to show uniform expansion slopes whereas the NBA cohort, a much larger cohort that captured early and late viral loads in most individuals, shows substantial variability in viral expansion slopes. In Figure 2D: the upslope seems extraordinarily rapid relative to other cohorts. I calculate a viral doubling time of roughly 1.5 hours. It would be helpful to understand how reliable of an estimate this is and also how much variability was observed among individuals.

We appreciate your detailed feedback on the estimated up-slope of viral dynamics. As the reviewer noted, the pattern differs from that observed in the NBA cohort, which may be due to their measurement of viral load from upper respiratory tract swabs. In our estimation, the mean and standard deviation of the doubling time (defined as $\ln 2 / (\beta T_{0pc}^{-1} - \delta)$) were 1.44 hours and 0.49 hours, respectively. Although direct validation of these values is challenging, several previous studies, including our own, have reported that viral loads in saliva increase more rapidly than in the upper respiratory tract swabs, reaching their peak sooner. Thus, we believe that our findings are consistent with those of previous studies. We revised our manuscript to discuss this point with additional references (page 20, lines 303-311).

(5) A key issue is that a lack of heterogeneity in the cohort may be driving a lack of differences between the groups. Table 1 shows that SpO2 values and lab values that all look normal. All infections were mild. This may make identifying biomarkers quite challenging.

Thank you for your comment regarding heterogeneity in the cohort. Although the NFV cohort was designed for COVID-19 patients who were either mild or asymptomatic, we have addressed this point and revised the manuscript to discuss it (page 21, lines 334-337).

(6) Figure 3A: many of the clinical variables such as basophil count, Cl, and protein have very low pre-test probability of correlating with virologic outcome.

Thank you for your comment regarding some clinical information we used in our study. We revised our manuscript to discuss this point (page 21, lines 337-338).

(7) A key omission appears to be microRNA from pre and early-infection time points. It would be helpful to understand whether microRNA levels at least differed between the two collection timepoints and whether certain microRNAs are dynamic during infection.

Thank you for your comment regarding the collection of micro-RNA data. As suggested by the reviewer, we compared micro-RNA levels between two time points using pairwise t-tests and Mann-Whitney U tests with FDR correction. As a result, no micro-RNA showed a statistically significant difference. This suggests that micro-RNA levels remain relatively stable during the course of infection, at least for mild or asymptomatic infection, and may therefore serve as a biomarker independent of sampling time. We have revised the manuscript to include this information (page 17, lines 259-262).

(8) The discussion could use a more thorough description of how viral kinetics differ in saliva versus nasal swabs and how this work complements other modeling studies in the field.

We appreciate the reviewer's thoughtful feedback. As suggested, we have added a discussion comparing our findings with studies that analyzed viral dynamics using nasal swabs, thereby highlighting the differences between viral dynamics in saliva and in the upper respiratory tract. To ensure a fair and rigorous comparison, we referred to studies that employed the

same mathematical model (i.e., Eqs.(1-2)). Accordingly, we revised the manuscript and included additional references (page 20, lines 303-311).

Furthermore, we clarified the significance of our study in two key aspects. First, it provides a detailed analysis of viral dynamics in saliva, reinforcing our previous findings from a single cohort by extending them across multiple cohorts. Second, this study uniquely examines whether viral dynamics in saliva can be directly predicted by exploring diverse clinical data and micro-RNAs. Notably, cohorts that have simultaneously collected and reported both viral load and a broad spectrum of clinical data from the same individuals, as in our study, are exceedingly rare. We revised the manuscript to clarify this point (page 20, lines 302-311).

(9) The most predictive potential variables of shedding heterogeneity which pertain to the innate and adaptive immune responses (virus-specific antibody and T cell levels) are not measured or modeled.

Thank you for your comment. We agree that antibody and T cell related markers may serve as the most powerful predictors, as supported by our own study [S. Miyamoto et al., PNAS (2023), ref. 24] as well as previous reports. While this point was already discussed in the manuscript, we have revised the text to make it more explicit (page 21, lines 327-328).

(10) I am curious whether the models infer different peak viral loads, duration, expansion, and clearance slopes between the 2 cohorts based on fitting to different infection stage data.

Thank you for your comment. We compared features between 2 cohorts as reviewer suggested. As a result, a statistically significant difference between the two cohorts (i.e., p-value ≤ 0.05 from the t-test) was observed only at the peak viral load, with overall trends being largely similar. At the peak, the mean value was $7.5 \log_{10}$ (copies/mL) in the Japan cohort and $8.1 \log_{10}$ (copies/mL) in the Illinois cohort, with variances of 0.88 and 0.87, respectively, indicating comparable variability.

Reviewer #2 (Public review)

Summary:

This study argues it has found that it has stratified viral kinetics for saliva specimens into three groups by the duration of "viral shedding"; the authors could not identify clinical data or microRNAs that correlate with these three groups.

Strengths:

The question of whether there is a stratification of viral kinetics is interesting.

Weaknesses:

The data underlying this work are not treated rigorously. The work in this manuscript is based on PCR data from two studies, with most of the data coming from a trial of nelfinavir (NFV) that showed no effect on the duration of SARS-CoV-2 PCR positivity. This study had no PCR data before symptom onset, and thus exclusively evaluated viral kinetics at or after peak viral loads. The second study is from the University of Illinois; this data set had sampling prior to infection, so has some ability to report the rate of "upswing." Problems in the analysis here include:

We are grateful to the reviewer for the constructive feedback, which has greatly enhanced the quality of our study. In response, we have carefully revised the manuscript to address all comments.

The PCR Ct data from each study is treated as equivalent and referred to as viral load, without any reports of calibration of platforms or across platforms. Can the authors provide calibration data and justify the direct comparison as well as the use of "viral load" rather than "Ct value"? Can the authors also explain on what basis they treat Ct values in the two studies as identical?

Thank you for your comment regarding description of viral load data. We recognized the lack of explanation for the integration of viral load data by reviewer's comment. We calculated viral load from Ct value using linear regression equations between Ct and viral load for each study's measurement method, respectively. We revised the manuscript to clarify this point in the section of Saliva viral load data in Methods.

The limit of detection for the NFV PCR data was unclear, so the authors assumed it was the same as the University of Illinois study. This seems a big assumption, as PCR platforms can differ substantially. Could the authors do sensitivity analyses around this assumption?

Thank you for your comment regarding the detection limit for viral load data. As reviewer suggested, we conducted sensitivity analysis for assumption of detection limit for the NFV dataset. Specifically, we performed data fitting in the same manner for two scenarios: when the detection limit of NFV PCR was lower ($0 \log_{10}$ copies/mL) or higher ($2 \log_{10}$ copies/mL) than that of the Illinois data ($1.08 \log_{10}$ copies/mL), and compared the results.

As a result, we obtained largely comparable viral dynamics in most cases (Supplementary Fig 6). When comparing the AIC values, we observed that the AIC for the same censoring threshold was 6836, whereas it increased to 7403 under the low censoring threshold and decreased to 6353 under the higher censoring threshold. However, this difference may be attributable to the varying number of data points treated as below the detection limit. Specifically, when the threshold is set higher, more data are treated as below the detection limit, which may result in a more favorable error calculation. To discuss this point, we have added a new figure (Supplementary Fig 6) and revised the manuscript accordingly (page 25, lines 415-418).

The authors refer to PCR positivity as viral shedding, but it is viral RNA detection (very different from shedding live/culturable virus, as shown in the Ke et al. paper). I suggest updating the language throughout the manuscript to be precise on this point.

We appreciate the reviewer's feedback regarding the terminology used for viral shedding. In response, we have revised all instances of "viral shedding" to "viral RNA detection" throughout the manuscript as suggested.

Eyeballing extended data in Figure 1, a number of the putative long-duration infections appear to be likely cases of viral RNA rebound (for examples, see S01-16 and S01-27). What happens if all the samples that look like rebound are reanalyzed to exclude the late PCR detectable time points that appear after negative PCRs?

We sincerely thank the reviewer for the valuable suggestion. In response, we established a criterion to remove data that appeared to exhibit rebound and subsequently performed data fitting

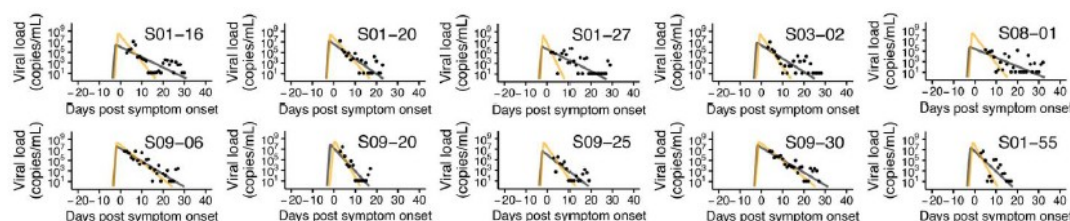
(see Author response image 1 below). The criterion was defined as: "any data that increase again after reaching the detection limit in two measurements are considered rebound and removed." As a result, 15 out of 144 cases were excluded due to insufficient usable data, leaving 129 cases for analysis. Using a single detection limit as the criterion would have

excluded too many data points, while defining the criterion solely based on the magnitude of increase made it difficult to establish an appropriate “threshold for increase.”

The fitting result indicates that the removal of rebound data may influence the fitting results; however, direct comparison of subsequent analyses, such as clustering, is challenging due to the reduced sample size. Moreover, the results can vary substantially depending on the criterion used to define rebound, and establishing a consistent standard remains difficult. Accordingly, we retained the current analysis and have added a discussion of rebound phenomena in the Discussion section as a limitation (page 22, lines 355-359). We once again sincerely appreciate the reviewer’s insightful and constructive suggestion.

Author response image 1.

Comparison of model fits before and after removing data suspected of rebound. Black dots represent observed measurements, and the black and yellow curves show the fitted viral dynamics for the full dataset and the dataset with rebound data removed, respectively.



There's no report of uncertainty in the model fits. Given the paucity of data for the upslope, there must be large uncertainty in the up-slope and likely in the peak, too, for the NFV data. This uncertainty is ignored in the subsequent analyses. This calls into question the efforts to stratify by the components of the viral kinetics. Could the authors please include analyses of uncertainty in their model fits and propagate this uncertainty through their analyses?

We sincerely appreciate the reviewer’s detailed feedback on model uncertainty. To address this point, we revised Extended Fig 1 (now renumbered as Supplementary Fig 1) to include 95% credible intervals computed using a bootstrap approach. In addition, to examine the potential impact of model uncertainty on stratified analyses, we reconstructed the distance matrix underlying stratification by incorporating feature uncertainty. Specifically, for each individual, we sampled viral dynamics within the credible interval and averaged the resulting feature, and build the distance matrix using it. We then compared this uncertainty-adjusted matrix with the original one using the Mantel test, which showed a strong correlation ($r = 0.72$, $p < 0.001$). Given this result, we did not replace the current stratification but revised the manuscript to provide this information through Result and Methods sections (page 11, lines 159-162 and page 28, lines 512-519). Once again, we are deeply grateful for this insightful comment.

The clinical data are reported as a mean across the course of an infection; presumably vital signs and blood test results vary substantially, too, over this duration, so taking a mean without considering the timing of the tests or the dynamics of their results is perplexing. I'm not sure what to recommend here, as the timing and variation in the acquisition of these clinical data are not clear, and I do not have a strong understanding of the basis for the hypothesis the authors are testing.

We appreciate the reviewers' feedback on the clinical data. We recognized that the manuscript lacked description of the handling of clinical data by your comment. In this

research, we focused on finding “early predictors” which could provide insight into viral shedding patterns. Thus, we used clinical data measured in the earliest time (date of admission) for each patient. Another reason is that the date of admission is the almost only time point at which complete clinical data without any missing values are available for all participants. We revised our manuscript to clarify this point (page 5, lines 90-95).

It's unclear why microRNAs matter. It would be helpful if the authors could provide more support for their claims that (1) microRNAs play such a substantial role in determining the kinetics of other viruses and (2) they play such an important role in modulating COVID-19 that it's worth exploring the impact of microRNAs on SARS-CoV-2 kinetics. A link to a single review paper seems insufficient justification. What strong experimental evidence is there to support this line of research?

We appreciate the reviewer’s comments regarding microRNA. Based on this feedback, we recognized the need to clarify our rationale for selecting microRNAs as the analyte. The primary reason was that our available specimens were saliva, and microRNAs are among the biomarkers that can be reliably measured in saliva. At the same time, previous studies have reported associations between microRNAs and various diseases, which led us to consider the potential relevance of microRNAs to viral dynamics, beyond their role as general health indicators. To better reflect this context, we have added supporting references (page 17, lines 240-243).

Reviewer #3 (Public review)

The article presents a comprehensive study on the stratification of viral shedding patterns in saliva among COVID-19 patients. The authors analyze longitudinal viral load data from 144 mildly symptomatic patients using a mathematical model, identifying three distinct groups based on the duration of viral shedding. Despite analyzing a wide range of clinical data and micro-RNA expression levels, the study could not find significant predictors for the stratified shedding patterns, highlighting the complexity of SARS-CoV-2 dynamics in saliva. The research underscores the need for identifying biomarkers to improve public health interventions and acknowledges several limitations, including the lack of consideration of recent variants, the sparsity of information before symptom onset, and the focus on symptomatic infections.

The manuscript is well-written, with the potential for enhanced clarity in explaining statistical methodologies. This work could inform public health strategies and diagnostic testing approaches. However, there is a thorough development of new statistical analysis needed, with major revisions to address the following points:

We sincerely appreciate the thoughtful feedback provided by Reviewer #3, particularly regarding our methodology. In response, we conducted additional analyses and revised the manuscript accordingly. Below, we address the reviewer’s comments point by point.

(1) Patient characterization & selection: Patient immunological status at inclusion (and if it was accessible at the time of infection) may be the strongest predictor for viral shedding in saliva. The authors state that the patients were not previously infected by SARS-COV-2. Was Anti-N antibody testing performed? Were other humoral measurements performed or did everything rely on declaration? From Figure 1A, I do not understand the rationale for excluding asymptomatic patients. Moreover, the mechanistic model can handle patients with only three observations, why are they not included? Finally, the 54 patients without clinical data can be used for the viral dynamics fitting and then discarded for the descriptive analysis. Excluding them can create a bias. All the discarded patients can help the virus dynamics analysis as it is a population approach. Please clarify. In Table 1 the absence of sex covariate is surprising.

We appreciate the detailed feedback from the reviewer regarding patient selection. We relied on the patient's self-declaration to determine the patient's history of COVID-19 infection and revised the manuscript to specify this (page 6, lines 83-84).

In parameter estimation, we used the date of symptom onset for each patient so that we establish a baseline of the time axis as clearly as possible, as we did in our previous works. Accordingly, asymptomatic patients who do not have information on the date of symptom onset were excluded from the analysis. Additionally, in the cohort we analyzed, for patients excluded due to limited number of observations (i.e., less than 3 points), most patients already had a viral load close to the detection limit at the time of the first measurement. This is due to the design of clinical trial, as if a negative result was obtained twice in a row, no further follow-up sampling was performed. These patients were excluded from the analysis because it hard to get reasonable fitting results. Also, we used 54 patients for the viral dynamics fitting and then only used the NFV cohort for clinical data analysis. We acknowledge that our description may have confused readers. We revised our manuscript to clarify these points regarding patient selecting for data fitting (page 6, lines 96-102, page 24, lines 406-407, and page 7, lines 410-412). In addition, we realized, thanks to the reviewer's comment, that gender information was missing in Table 1. We appreciate this observation and have revised the table to include gender (we used gender in our analysis).

(2) Exact study timeline for explanatory covariates: I understand the idea of finding « early predictors » of long-lasting viral shedding. I believe it is key and a great question. However, some samples (Figure 4A) seem to be taken at the end of the viral shedding. I am not sure it is really easier to micro-RNA saliva samples than a PCR. So I need to be better convinced of the impact of the possible findings. Generally, the timeline of explanatory covariate is not described in a satisfactory manner in the actual manuscript. Also, the evaluation and inclusion of the daily symptoms in the analysis are unclear to me.

We appreciate the reviewer's feedback regarding the collection of explanatory variables. As noted, of the two microRNA samples collected from each patient, one was obtained near the end of viral shedding. This was intended to examine potential differences in microRNA levels between the early and late phases of infection. No significant differences were observed between the two time points, and using microRNA from either phase alone or both together did not substantially affect predictive accuracy for stratified groups. Furthermore, microRNA collection was motivated primarily by the expectation that it would be more sensitive to immune responses, rather than by ease of sampling. We have revised the manuscript to clarify these points regarding microRNA (page 17, lines 243-245 and 259-262).

Furthermore, as suggested by the reviewer, we have also strengthened the explanation regarding the collection schedule of clinical information and the use of daily symptoms in the analysis (page 6, lines 90-95, page 14, lines 218-220.).

(3) Early Trajectory Differentiation: The model struggles to differentiate between patients' viral load trajectories in the early phase, with overlapping slopes and indistinguishable viral load peaks observed in Figures 2B, 2C, and 2D. The question arises whether this issue stems from the data, the nature of Covid-19, or the model itself. The authors discuss the scarcity of pre-symptom data, primarily relying on Illinois patients who underwent testing before symptom onset. This contrasts earlier statements on pages 5-6 & 23, where they claim the data captures the full infection dynamics, suggesting sufficient early data for pre-symptom kinetics estimation. The authors need to provide detailed information on the number or timing of patient sample collections during each period.

Thank you for the reviewer's thoughtful comments. The model used in this study [Eqs.(1-2)] has been employed in numerous prior studies and has successfully identified viral dynamics at the individual level. In this context, we interpret the rapid viral increase observed across participants as attributable to characteristics of SARS-CoV-2 in saliva, an interpretation that has also been reported by multiple previous studies. We have added the relevant references and strengthened the corresponding discussion in the manuscript (page 20, lines 303-311).

We acknowledge that our explanation of how the complementary relationship between the two cohorts contributes to capturing infection dynamics was not sufficiently clear. As described in the manuscript, the Illinois cohort provides pre-symptomatic data, whereas the NFV cohort offers abundant end-phase data, thereby compensating for each other's missing phases. By jointly analyzing the two cohorts with a nonlinear mixed-effects model, we estimated viral dynamics at the individual-level. This approach first estimates population-level parameters (fixed effects) using data from all participants and then incorporates random effects to account for individual variability, yielding the most plausible parameter values.

Thus, even when early-phase data are lacking in the NFV cohort, information from the Illinois cohort allows us to infer most reasonable dynamics, and the reverse holds true for the end phase. In this context, we argued that combining the two cohorts enables mathematical modeling to capture infection dynamics at the individual level. Recognizing that our earlier description could be misleading, we have carefully reinforced the relevant description (page 27, lines 472-483). In addition, as suggested by the reviewer, we have added information on the number of data samples available for each phase in both cohorts (page 7, lines 106-109).

(4) Conditioning on the future: Conditioning on the future in statistics refers to the problematic situation where an analysis inadvertently relies on information that would not have been available at the time decisions were made or data were collected. This seems to be the case when the authors create micro-RNA data (Figure 4A). First, when the sampling times are is something that needs to be clarified by the authors (for clinical outcomes as well). Second, proper causal inference relies on the assumption that the cause precedes the effect. This conditioning on the future may result in overestimating the model's accuracy. This happens because the model has been exposed to the outcome it's supposed to predict. This could question the - already weak - relation with mir-1846 level.

We appreciate the reviewer's detailed feedback. As noted in Reply to Comments 2, we collected micro-RNA samples at two time points, near the peak of infection dynamics and at the end stage, and found no significant differences between them. This suggests that micro-RNA levels are not substantially affected by sampling time. Indeed, analyses conducted using samples from the peak, late stage, or both yielded nearly identical results in relation to infection dynamics. To clarify this point, we revised the manuscript by integrating this explanation with our response in Reply to Comments 2 (page 17, lines 259-262). In addition, now we also revised manuscript to clarify sampling times of clinical information and micro-RNA (page 6, lines 90-95).

(5) Mathematical Model Choice Justification and Performance: The paper lacks mention of the practical identifiability of the model (especially for tau regarding the lack of early data information). Moreover, it is expected that the immune effector model will be more useful at the beginning of the infection (for which data are the more parsimonious). Please provide AIC for comparison, saying that they have "equal performance" is not enough. Can you provide at least in a point-by-point response the VPC & convergence assessments?

We appreciate the reviewer's detailed feedback regarding the mathematical model. We acknowledge the potential concern regarding the practical identifiability of tau (incubation period), particularly given the limited early-phase data. In our analysis, however, the nonlinear mixed-effects model yielded a population-level estimate of 4.13 days, which is similar with previously reported incubation periods for COVID-19. This concordance suggests that our estimate of tau is reasonable despite the scarcity of early data.

For model comparison, first, we have added information on the AIC of the two models to the manuscript as suggested by the reviewer (page 10, lines 130-135). One point we would like to emphasize is that we adopted a simple target cell-limited model in this study, aiming to focus on reconstruction of viral dynamics and stratification of shedding patterns rather than exploring the mechanism of viral infection in detail. Nevertheless, we believe that the target cell-limited model provides reasonable reconstructed viral dynamics as it has been used in many previous studies. We revised manuscript to clarify this (page 10, lines 135-144).

Furthermore, as suggested, we have added the VPC and convergence assessment results for both models, together with explanatory text, to the manuscript (Supplementary Fig 2, Supplementary Fig 3, and page 10, lines 130-135). In the VPC, the observed 5th, 50th, and 95th percentiles were generally within the corresponding simulated prediction intervals across most time points. Although minor deviations were noted in certain intervals, the overall distribution of the observed data was well captured by the models, supporting their predictive performance (Supplementary Fig 2). In addition, the log-likelihood and SAEM parameter trajectories stabilized after the burn-in phase, confirming appropriate convergence (Supplementary Fig 3).

(6) Selected features of viral shedding: I wonder to what extent the viral shedding area under the curve (AUC) and normalized AUC should be added as selected features.

We sincerely appreciate the reviewer's valuable suggestion regarding the inclusion of additional features. Following this recommendation, we considered AUC (or normalized AUC) as an additional feature when constructing the distance matrix used for stratification. We then evaluated the similarity between the resulting distance matrix and the original one using the Mantel test, which showed a very high correlation ($r = 0.92$, $p < 0.001$). This indicates that incorporating AUC as an additional feature does not substantially alter the distance matrix. Accordingly, we have decided to retain the current stratification analysis, and we sincerely thank the reviewer once again for this interesting suggestion.

(7) Two-step nature of the analysis: First you fit a mechanistic model, then you use the predictions of this model to perform clustering and prediction of groups (unsupervised then supervised). Thus you do not propagate the uncertainty intrinsic to your first estimation through the second step, ie. all the viral load selected features actually have a confidence bound which is ignored. Did you consider a one-step analysis in which your covariates of interest play a direct role in the parameters of the mechanistic model as covariates? To pursue this type of analysis SCM (Johnson et al. Pharm. Res. 1998), COSSAC (Ayrat et al. 2021 CPT PsP), or SAMBA (Prague et al. CPT PsP 2021) methods can be used. Did you consider sampling on the posterior distribution rather than using EBE to avoid shrinkage?

Thank you for the reviewer's detailed suggestions regarding our analysis. We agree that the current approach does not adequately account for the impact of uncertainty in viral dynamics on the stratified analyses. As a first step, we have revised Extended Data Fig 1 (now renumbered as Supplementary Fig 1) to include 95% credible intervals computed using a bootstrap approach, to present the model-fitting uncertainty more explicitly. Then, to examine the potential impact of model uncertainty on stratified analyses, we reconstructed

the distance matrix underlying stratification by incorporating feature uncertainty. Specifically, for each individual, we sampled viral dynamics within the credible interval and averaged the resulting feature, and build the distance matrix using it. We then compared this uncertainty-adjusted matrix with the original one using the Mantel test, which showed a strong correlation ($r = 0.72$, $p < 0.001$). Given this result, we did not replace the current stratification but revised the manuscript to provide this information (page 11, lines 159-162 and page 28, 512-519).

Furthermore, we carefully considered the reviewer's proposed one-step analysis. However, implementation was constrained by data-fitting limitations. Concretely, clinical information is available only in the NFV cohort. Thus, if these variables are to be entered directly as covariates on the parameters, the Illinois cohort cannot be included in the data-fitting process. Yet the NFV cohort lacks any pre-symptomatic observations, so fitting the model to that cohort alone does not permit a reasonable (well-identified/robust) fitting result. While we were unable to implement the suggestion under the current data constraints, we sincerely appreciate the reviewer's thoughtful and stimulating proposal.

(8) Need for advanced statistical methods: The analysis is characterized by a lack of power. This can indeed come from the sample size that is characterized by the number of data available in the study. However, I believe the power could be increased using more advanced statistical methods. At least it is worth a try. First considering the unsupervised clustering, summarizing the viral shedding trajectories with features collapses longitudinal information. I wonder if the R package « LongituRF » (and associated method) could help, see Capitaine et al. 2020 SMMR. Another interesting tool to investigate could be latent class models R package « lcmm » (and associated method), see ProustLima et al. 2017 J. Stat. Softwares. But the latter may be more far-reaching.

Thank you for the reviewer's thoughtful suggestions regarding our unsupervised clustering approach. The R package "LongituRF" is designed for supervised analysis, requiring a target outcome to guide the calculation of distances between individuals (i.e., between viral dynamics). In our study, however, the goal was purely unsupervised clustering, without any outcome variable, making direct application of "LongituRF" challenging.

Our current approach (summarizing each dynamic into several interpretable features and then using Random Forest proximities) allows us to construct a distance matrix in an unsupervised manner. Here, the Random Forest is applied in "proximity mode," focusing on how often dynamics are grouped together in the trees, independent of any target variable. This provides a practical and principled way to capture overall patterns of dynamics while keeping the analysis fully unsupervised.

Regarding the suggestion to use latent class mixed models (R package "lcmm"), we also considered this approach. In our dataset, each subject has dense longitudinal measurements, and at many time points, trajectories are very similar across subjects, resulting in minimal inter-individual differences. Consequently, fitting multi-class latent class mixed models ($ng \geq 2$) with random effects or mixture terms is numerically unstable, often producing errors such as non-positive definite covariance matrices or failure to generate valid initial values. Although one could consider using only the time points with the largest differences, this effectively reduces the analysis to a feature-based summary of dynamics. Such an approach closely resembles our current method and contradicts the goal of clustering based on full longitudinal information.

Taken together, although we acknowledge that incorporating more longitudinal information is important, we believe that our current approach provides a practical, stable, and informative solution for capturing heterogeneity in viral dynamics. We would like to once again express our sincere gratitude to the reviewer for this insightful suggestion.

(9) Study intrinsic limitation: All the results cannot be extended to asymptomatic patients and patients infected with recent VOCs. It definitively limits the impact of results and their applicability to public health. However, for me, the novelty of the data analysis techniques used should also be taken into consideration.

We appreciate your positive evaluation of our research approach and acknowledge that, as noted in the Discussion section as our first limitation, our analysis may not provide valid insights into recent VOCs or all populations, including asymptomatic individuals. Nonetheless, we believe it is novel that we extensively investigated the relationship between viral shedding patterns in saliva and a wide range of clinical and micro-RNA data. Our findings contribute to a deeper and more quantitative understanding of heterogeneity in viral dynamics, particularly in saliva samples. To discuss this point, we revised our manuscript (page 22, lines 364-368).

Strengths are:

Unique data and comprehensive analysis.

Novel results on viral shedding.

Weaknesses are:

Limitation of study design.

The need for advanced statistical methodology.

Reviewer #1 (Recommendations For The Authors):

Line 8: In the abstract, it would be helpful to state how stratification occurred.

We thank the reviewer for the feedback, and have revised the manuscript accordingly (page 2, lines 8-11).

Line 31 and discussion: It is important to mention the challenges of using saliva as a specimen type for lab personnel.

We thank the reviewer for the feedback, and have revised the manuscript accordingly (page 3, lines 36-41).

Line 35: change to "upper respiratory tract".

We thank the reviewer for the feedback, and have revised the manuscript accordingly (page 3, line 35).

Line 37: "Saliva" is not a tissue. Please hazard a guess as to which tissue is responsible for saliva shedding and if it overlaps with oral and nasal swabs.

We thank the reviewer for the feedback, and have revised the manuscript accordingly (page 3, lines 42-45).

Line 42, 68: Please explain how understanding saliva shedding dynamics would impact isolation & screening, diagnostics, and treatments. This is not immediately intuitive to me.

We thank the reviewer for the feedback, and have revised the manuscript accordingly (page 3, lines 48-50).

Line 50: It would be helpful to explain why shedding duration is the best stratification variable.

We thank the reviewer for the feedback. We acknowledge that our wording was ambiguous. The clear differences in the viral dynamics patterns pertain to findings observed following the stratification, and we have revised the manuscript to make this explicit (page 4, lines 59-61).

Line 71: Dates should be listed for these studies.

We thank the reviewer for the feedback, and have revised the manuscript accordingly (page 6, lines 85-86).

Reviewer #2 (Recommendations For The Authors):

Please make all code and data available for replication of the analyses.

We appreciate the suggestion. Due to ethical considerations, it is not possible to make all data and code publicly available. We have clearly stated in the manuscript about it (Data availability section in Methods).

Reviewer #3 (Recommendations For The Authors):

Here are minor comments / technical details:

(1) Figure 1B is difficult to understand.

Thank you for the comment. We updated Fig 1B to incorporate more information to aid interpretation.

(2) Did you analyse viral load or the log10 of viral load? The latter is more common. You should consider it. SI Figure 1 please plot in log10 and use a different point shape for censored data. The file quality of this figure should be improved. State in the material and methods if SE with moonlit are computed with linearization or importance sampling.

Thank you for the comment. We conducted our analyses using log10-transformed viral load. Also, we revised Supplementary Fig 1 (now renumbered as Supplementary Fig 4) as suggested. We also added Supplementary Fig 3 and clarified in the Methods that standard errors (SE) were obtained in Monolix from the Fisher information matrix using the linearization method (page 28, lines 498-499).

(3) Table 1 and Figure 3A could be collapsed.

Thank you for the comment, and we carefully considered this suggestion. Table 1 summarizes clinical variables by category, whereas Fig 3A visualizes them ordered by p-value of statistical analysis. Collapsing these into a single table would make it difficult to apprehend both the categorical summaries and the statistical ranking at a glance, thereby reducing readability. We therefore decided to retain the current layout. We appreciate the constructive feedback again.

(4) Figure 3 legend could be clarified to understand what is 3B and 3C.

We thank the reviewer for the feedback and have reinforced the description accordingly.

| (5) *Why use AIC instead of BICc?*

Thank you for your comment. We also think BICc is a reasonable alternative. However, because our objective is predictive adequacy (reconstruction of viral dynamics), we judged AIC more appropriate. In NLMEM settings, the effective sample size required by BICc is ambiguous, making the penalty somewhat arbitrary. Moreover, since the two models reconstruct very similar dynamics, our conclusions are not sensitive to the choice of criterion.

| (6) *Bibliography. Most articles are with et al. (which is not standard) and some are with an extended list of names. Provide DOI for all.*

We thank the reviewer for the feedback, and have revised the manuscript accordingly.

| (7) *Extended Table 1&2 - maybe provide a color code to better highlight some lower p-values (if you find any interesting).*

We thank the reviewer for the feedback. Since no clinical information and micro-RNAs other than mir-1846 showed low p-values, we highlighted only mir-1846 with color to make it easier to locate.

| (8) *Please make the replication code available.*

We appreciate the suggestion. Due to ethical considerations, it is not possible to make all data and code publicly available. We have clearly stated in the manuscript about it (Data availability section in Methods).

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